Theoretical Foundations

VLEO Slew Maneuver

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1 Coordinate Transformations

1.1 Quaternion to Direction Cosine Matrix

A quaternion $\boldsymbol{q} = [q_0\ q_1\ q_2\ q_3]^\mathsf{T}$ using the scalar-first convention can be converted to a direction cosine matrix through

$$\mathbf{C}(\boldsymbol{q}) = \begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1 q_2 + q_0 q_3) & 2(q_1 q_3 - q_0 q_2) \\ 2(q_1 q_2 - q_0 q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2 q_3 + q_0 q_1) \\ 2(q_1 q_3 + q_0 q_2) & 2(q_2 q_3 - q_0 q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix}. \tag{1}$$

Source. Wertz, J. R., *Spacecraft Attitude Determination and Control*, 1978, Eq. 12-12.

Implemented in. Aerospace Toolbox `quat2dcm`, used in `src/Sat_template.m`, `src/sat_template_gui.m`, `src/gui/AttitudeAnimator.m`, `src/gui/AttitudeAnimatorGUI.m`, and `src/computeAeroForces.m`.

1.2 Direction Cosine Matrix to Quaternion

The inverse transformation uses Shepperd's method for numerical stability:

1. Compute the matrix trace and diagonal elements.
2. Find the maximum of $\{\mathrm{tr}(\mathbf{C}), C_{11}, C_{22}, C_{33}\}$.
3. Use the corresponding extraction formula to avoid division by a small number.

Source. Shepperd, S. W., "Quaternion from Rotation Matrix," *Journal of Guidance and Control*, Vol. 1, No. 3, 1978.

Implemented in. Aerospace Toolbox `dcm2quat`, used in `src/Reference_Trajectory_Calculation.m` and `src/gui/runSimulation.m`.

1.3 Quaternion and Principal Axis-Angle

Quaternion from axis-angle:

$$q_0 = \cos\left(\frac{\theta}{2}\right), \qquad q_1 = e_1 \sin\left(\frac{\theta}{2}\right), \qquad q_2 = e_2 \sin\left(\frac{\theta}{2}\right), \qquad q_3 = e_3 \sin\left(\frac{\theta}{2}\right). \tag{2}$$

Here $\boldsymbol{e} = [e_1\ e_2\ e_3]^\mathsf{T}$ is the unit rotation axis and θ is the rotation angle.

Axis-angle from quaternion:

$$\theta = 2\cos^{-1}(q_0), \tag{3}$$

with

$$
\boldsymbol{e} = \begin{cases} \dfrac{1}{\sin(\theta/2)}[q_1\ q_2\ q_3]^{\mathsf{T}}, & \theta \neq 0, \\ [1\ 0\ 0]^{\mathsf{T}}, & \theta = 0. \end{cases} \tag{4}
$$

Source. Euler's rotation theorem; Kuipers, J., *Quaternions and Rotation Sequences*, 1999.

Implemented in. No current `src/` implementation. If principal-axis-angle support is needed again, add it back as a documented custom repository function.

1.4 Fundamental Rotation Matrices

Elementary rotations about the principal axes are

$$
\mathbf{R}_1(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix}, \tag{5}
$$

$$
\mathbf{R}_2(\theta) = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}, \tag{6}
$$

$$
\mathbf{R}_3(\theta) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{7}
$$

Source. Schaub, H. and Junkins, J., *Analytical Mechanics of Space Systems*, 4th ed., 2018, Ch. 3.

Implemented in. No dedicated `src/` helper. Current `src/` code uses Aerospace Toolbox rotation and conversion utilities instead of a local elementary-rotation wrapper.

2 Orbital Mechanics

2.1 Orbital Elements from Position and Velocity

Given inertial position $\boldsymbol{r}$ and velocity $\boldsymbol{v}$, the classical orbital elements are computed through

1. Specific angular momentum:
$$
\boldsymbol{h} = \boldsymbol{r} \times \boldsymbol{v}. \tag{8}
$$

2. Node vector:
$$
\boldsymbol{n} = [0\ 0\ 1]^{\mathsf{T}} \times \boldsymbol{h}. \tag{9}
$$

3. Eccentricity vector:
$$
\boldsymbol{e} = \frac{\left(\|\boldsymbol{v}\|^2 - \mu/\|\boldsymbol{r}\|\right)\boldsymbol{r} - (\boldsymbol{r}\cdot\boldsymbol{v})\boldsymbol{v}}{\mu}. \tag{10}
$$

4. Specific orbital energy:
$$\epsilon = \frac{\|\boldsymbol{v}\|^2}{2} - \frac{\mu}{\|\boldsymbol{r}\|}. \tag{11}$$
5. Semi-major axis:
$$a = -\frac{\mu}{2\epsilon}. \tag{12}$$
6. Inclination:
$$i = \cos^{-1}\left(\frac{h_z}{\|\boldsymbol{h}\|}\right). \tag{13}$$
7. Right ascension of the ascending node:
$$\Omega = \cos^{-1}\left(\frac{n_x}{\|\boldsymbol{n}\|}\right), \tag{14}$$
with the quadrant selected from the sign of n_y.
8. Argument of periapsis:
$$\omega = \cos^{-1}\left(\frac{\boldsymbol{n}\cdot\boldsymbol{e}}{\|\boldsymbol{n}\|\,\|\boldsymbol{e}\|}\right), \tag{15}$$
with the quadrant selected from the sign of e_z.
9. True anomaly:
$$f = \cos^{-1}\left(\frac{\boldsymbol{e}\cdot\boldsymbol{r}}{\|\boldsymbol{e}\|\,\|\boldsymbol{r}\|}\right), \tag{16}$$
with the quadrant selected from the sign of $\boldsymbol{r}\cdot\boldsymbol{v}$.

Special cases.

- Circular orbits ($e \approx 0$): use the argument of latitude $u = \omega + f$.
- Equatorial orbits ($i \approx 0$): use the longitude of periapsis $\varpi = \Omega + \omega$.

Source. Vallado, D. A., *Fundamentals of Astrodynamics and Applications*, 4th ed., 2013, Algorithm 9.

Implemented in. Aerospace Toolbox `ijk2keplerian` when state-to-elements conversion is needed.

2.2 Position and Velocity from Orbital Elements

First compute the position and velocity in the perifocal frame:

$$\boldsymbol{r}_{\mathrm{PQW}} = \frac{a(1-e^2)}{1+e\cos f}\begin{bmatrix}\cos f\\ \sin f\\ 0\end{bmatrix}, \tag{17}$$
$$\boldsymbol{v}_{\mathrm{PQW}} = \sqrt{\frac{\mu}{a(1-e^2)}}\begin{bmatrix}-\sin f\\ e+\cos f\\ 0\end{bmatrix}. \tag{18}$$

Then transform to the ECI frame with

$$\begin{aligned}
\mathbf{R} &= \mathbf{R}_3(-\Omega)\mathbf{R}_1(-i)\mathbf{R}_3(-\omega), && (19)\\
\boldsymbol{r}_{\mathrm{ECI}} &= \mathbf{R}\boldsymbol{r}_{\mathrm{PQW}}, && (20)\\
\boldsymbol{v}_{\mathrm{ECI}} &= \mathbf{R}\boldsymbol{v}_{\mathrm{PQW}}. && (21)
\end{aligned}$$

Source. Vallado, D. A., *Fundamentals of Astrodynamics and Applications*, 4th ed., 2013, Algorithm 10.

Implemented in. Aerospace Toolbox `keplerian2ijk`, used in `src/Reference_Trajectory_Calculation.m`, `src/gui/runSimulation.m`, and `src/gui/AttitudeAnimator.m`.

3 Attitude Dynamics

3.1 Quaternion Kinematics

The time derivative of the quaternion representing body orientation is

$$\dot{\boldsymbol{q}} = \frac{1}{2}\boldsymbol{\Omega}(\boldsymbol{\omega})\boldsymbol{q}, \tag{22}$$

where the quaternion rate matrix is

$$\boldsymbol{\Omega}(\boldsymbol{\omega}) = \begin{bmatrix} 0 & -\omega_1 & -\omega_2 & -\omega_3 \\ \omega_1 & 0 & \omega_3 & -\omega_2 \\ \omega_2 & -\omega_3 & 0 & \omega_1 \\ \omega_3 & \omega_2 & -\omega_1 & 0 \end{bmatrix}. \tag{23}$$

Here $\boldsymbol{\omega} = [\omega_1\ \omega_2\ \omega_3]^\mathsf{T}$ is the angular velocity of the body frame relative to the inertial frame, expressed in body coordinates.

Source. Wertz, J. R., *Spacecraft Attitude Determination and Control*, 1978, Section 12.1.

Implemented in. `src/Sat_template.m`.

3.2 Euler's Rotational Equations

For a rigid body with inertia tensor $\mathbf{J}$,

$$\mathbf{J}\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega}) + \boldsymbol{T}_{\mathrm{ext}}. \tag{24}$$

For a diagonal inertia tensor $\mathbf{J} = \mathrm{diag}([J_x, J_y, J_z])$,

$$J_x\dot{\omega}_x = (J_y - J_z)\omega_y\omega_z + T_x, \tag{25}$$
$$J_y\dot{\omega}_y = (J_z - J_x)\omega_z\omega_x + T_y, \tag{26}$$
$$J_z\dot{\omega}_z = (J_x - J_y)\omega_x\omega_y + T_z. \tag{27}$$

Source. Schaub, H. and Junkins, J., *Analytical Mechanics of Space Systems*, 4th ed., 2018, Ch. 4.

Implemented in. `src/Sat_template.m`.

3.3 PD Attitude Control Law

Quaternion-based feedback control is written as

$$\boldsymbol{T}_{\text{control}} = -\mathbf{K}_p \boldsymbol{q}_{\text{error},v} - \mathbf{K}_d \boldsymbol{\omega}_{\text{error}}, \tag{28}$$

where

- $\boldsymbol{q}_{\text{error},v}$ is the vector part of the attitude error quaternion,
- $\boldsymbol{\omega}_{\text{error}}$ is the angular velocity error,
- $\mathbf{K}_p$ is the proportional gain matrix, and
- $\mathbf{K}_d$ is the derivative gain matrix.

Source. Wie, B., *Space Vehicle Dynamics and Control*, 2nd ed., 2008, Ch. 7.

Implemented in. `src/controllers/control_torques.m`.

4 Perturbations

4.1 J2 Perturbation (Earth Oblateness)

The acceleration due to Earth's J_2 zonal harmonic is

$$\boldsymbol{a}_{J_2} = \frac{3}{2} J_2 \frac{\mu}{r^2} \left(\frac{R_E}{r}\right)^2 \begin{bmatrix} \left(\frac{x}{r}\right)\left(5\left(\frac{z}{r}\right)^2 - 1\right) \\ \left(\frac{y}{r}\right)\left(5\left(\frac{z}{r}\right)^2 - 1\right) \\ \left(\frac{z}{r}\right)\left(5\left(\frac{z}{r}\right)^2 - 3\right) \end{bmatrix}. \tag{29}$$

Here $J_2 = 1.08263 \times 10^{-3}$ is dimensionless, $R_E = 6378137\,\text{m}$ is the Earth's equatorial radius, $\mu = 3.986004418 \times 10^{14}\,\text{m}^3/\text{s}^2$ is the Earth gravitational parameter, and $r = \|\boldsymbol{r}\|$ is the distance from the Earth's center.

Source. Vallado, D. A., *Fundamentals of Astrodynamics and Applications*, 4th ed., 2013, Eq. 8-25.

Implemented in. Aerospace Toolbox `gravityzonal`, used in `src/Sat_template.m` and `src/sat_template_gui.`

4.2 J3 Perturbation

The acceleration due to Earth's J_3 zonal harmonic is

$$\boldsymbol{a}_{J_3} = \frac{1}{2} J_3 \frac{\mu}{r^2} \left(\frac{R_E}{r}\right)^3 \begin{bmatrix} \left(\frac{x}{r}\right)\left(10\left(\frac{z}{r}\right)^3 - 15\left(\frac{z}{r}\right)\right) \\ \left(\frac{y}{r}\right)\left(10\left(\frac{z}{r}\right)^3 - 15\left(\frac{z}{r}\right)\right) \\ 4\left(\frac{z}{r}\right)^3 - 3\left(\frac{z}{r}\right) \end{bmatrix}, \tag{30}$$

with $J_3 = -2.5327 \times 10^{-6}$.

Source. Vallado, D. A., *Fundamentals of Astrodynamics and Applications*, 4th ed., 2013, Eq. 8-26.

Implemented in. Aerospace Toolbox `gravityzonal`, used in `src/Sat_template.m` and `src/sat_template_gui.`

4.3 Atmospheric Drag in VLEO

The drag acceleration model is

$$\boldsymbol{a}_{\text{drag}} = -\frac{1}{2} \rho C_d \frac{A}{m} \|\boldsymbol{v}_{\text{rel}}\| \boldsymbol{v}_{\text{rel}}. \tag{31}$$

Here ρ is the atmospheric density, C_d is the drag coefficient, A is the cross-sectional area, m is the spacecraft mass, and $\boldsymbol{v}_{\text{rel}}$ is the velocity relative to the atmosphere.

Atmospheric models.

- Exponential atmosphere (simple)
- NRLMSISE-00 (high fidelity)
- JB2008 (operational)

Source. Vallado, D. A., *Fundamentals of Astrodynamics and Applications*, 4th ed., 2013, Ch. 8.6.

Implemented in. `src/computeAeroForces.m`, which evaluates NRLMSISE-00 using geodetic coordinates resolved from either direct latitude/longitude inputs or the spacecraft ECI position and simulation time. The resulting forces and moments are applied in `src/Sat_template.m`, `src/sat_template_gui.m`, and `src/gui/runSimulation.m`.

4.4 Solar Radiation Pressure

The solar radiation pressure acceleration model is

$$\boldsymbol{a}_{\text{SRP}} = -P_{\text{sr}} C_r \frac{A}{m} \frac{\boldsymbol{r}_{\text{sun}}}{\|\boldsymbol{r}_{\text{sun}}\|}. \tag{32}$$

Here $P_{\text{sr}} \approx 4.56 \times 10^{-6}\,\text{N/m}^2$ at 1 AU, C_r is the reflectivity coefficient, A is the illuminated area, and $\boldsymbol{r}_{\text{sun}}$ is the position vector from the spacecraft to the Sun.

Source. Montenbruck, O. and Gill, E., *Satellite Orbits*, 2000, Ch. 3.4.

Implemented in. `src/computeAeroForces.m`, which computes panel-wise solar coefficients and returns total solar force and moment contributions for the active spacecraft mesh.

5 Propulsion

5.1 Ion Thruster Fundamentals

The thrust relation is

$$F = \dot{m} v_e = \dot{m} g_0 I_{sp}, \tag{33}$$

where F is thrust, $\dot{m}$ is mass flow rate, v_e is exhaust velocity, $g_0 = 9.80665\,\text{m/s}^2$ is standard gravity, and I_{sp} is specific impulse.

The corresponding power-thrust relation is

$$P = \frac{1}{2} \dot{m} v_e^2 \frac{1}{\eta}, \tag{34}$$

where η is the thruster efficiency.

Source. Goebel, D. and Katz, I., *Fundamentals of Electric Propulsion: Ion and Hall Thrusters*, JPL, 2008.

Implemented in. TBD.

5.2 Slew Maneuver Analysis

For a rest-to-rest slew maneuver of angle θ, the minimum-time bang-bang control result is

$$t_{\min} = 2\sqrt{\frac{\theta J}{T_{\max}}}, \tag{35}$$

and a smooth minimum-energy form is written as

$$E_{\min} = \frac{4}{3} \frac{J\theta^2}{t^3}. \tag{36}$$

Here J is the moment of inertia about the rotation axis and $T_{\max}$ is the maximum available torque.

Source. Wie, B., *Space Vehicle Dynamics and Control*, 2nd ed., 2008, Ch. 7.

Implemented in. TBD.

6 References

6.1 Textbooks

1. Vallado, D. A. (2013). *Fundamentals of Astrodynamics and Applications*, 4th ed. Microcosm Press.

2. Wertz, J. R. (1978). *Spacecraft Attitude Determination and Control*. Kluwer Academic Publishers.

3. Schaub, H. and Junkins, J. (2018). *Analytical Mechanics of Space Systems*, 4th ed. AIAA.

4. Wie, B. (2008). *Space Vehicle Dynamics and Control*, 2nd ed. AIAA.

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6. Kuipers, J. (1999). *Quaternions and Rotation Sequences*. Princeton University Press.

7. Goebel, D. and Katz, I. (2008). *Fundamentals of Electric Propulsion: Ion and Hall Thrusters*. JPL/Wiley.

6.2 Papers

8. Shepperd, S. W. (1978). "Quaternion from Rotation Matrix." *Journal of Guidance and Control*, Vol. 1, No. 3.

6.3 Standards

9. IERS Conventions (2010). IERS Technical Note No. 36.

10. WGS84 (2014). NGA.STND.0036_1.0.0_WGS84.

Document History

Date	Author	Changes
2025-12-30	Team	Initial template with existing implementations